



Training Self-Aware LLMs Against Prompt Injection via Preference Optimization

Project Code: CN.NC.SV.25_08

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Build the first model-level Prompt Injection (PI) defense mechanism for the Vietnamese language

-17.75 pp

**Average Attack
Success Rate
(ASR) Reduction**

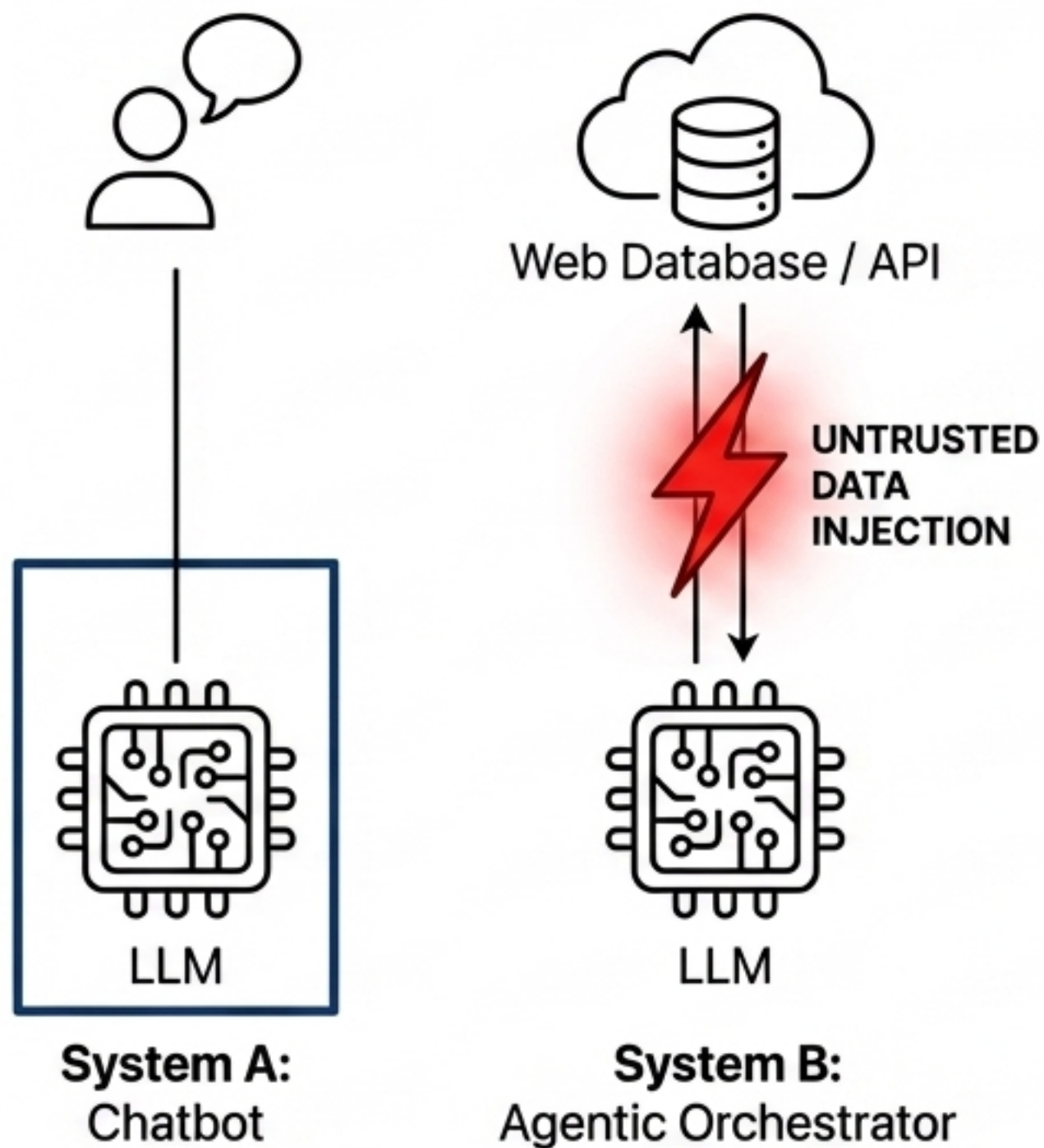
-0.03 pp

**Zero Alignment Tax.
Maintained MMLU
score with almost
imperceptible
degradation.**

26,819

**Novel bilingual
(VI-EN) preference
samples created
and open-sourced.**

The Paradigm Shift: From Chatbots to Agentic Orchestrators



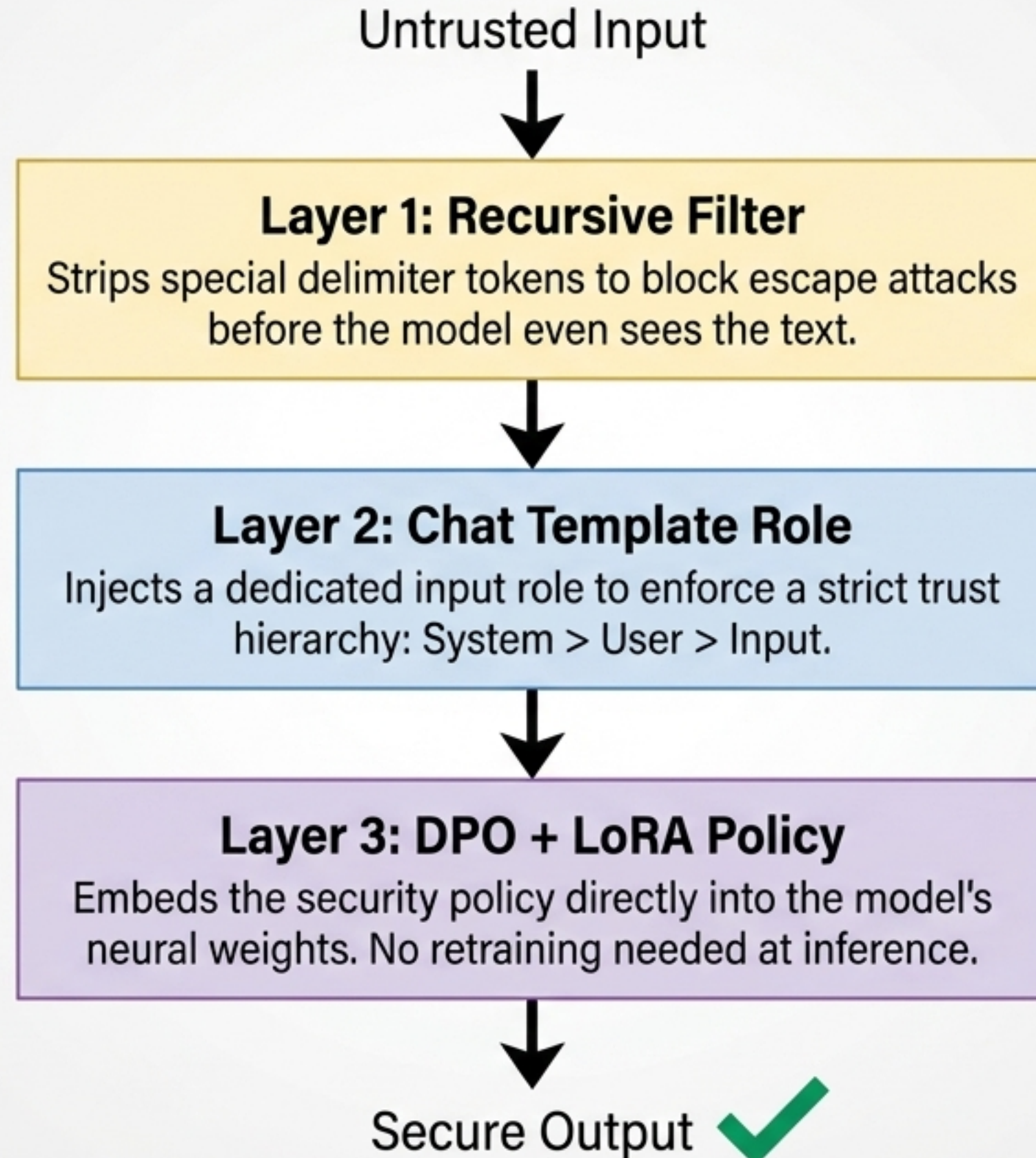
LLMs process system instructions and untrusted external data in the exact same token sequence.

#1 Security Threat: OWASP Top 10 for LLM Applications 2025.

The Gap (GAP-01): Absolute absence of open-source, model-level defense for Vietnamese language systems.

[15-Second
Video Demo:
Live Prompt
Injection
Hijacking]

The 3-Layer Defense Architecture



Mathematical Foundation: Preference Optimization

Safe Response

(Chosen)

Hijacked Response

(Rejected)

Maximizes
Probability
Margin

RPO

cDPO (Label Smoothing):

Prevents over-refusal &
overconfidence



Utility (MMLU)

Direct Preference Optimization (DPO):

Optimizes policy directly from preference data without a separate reward model.

Rejection Pair Optimization (RPO):

Injects Supervised Fine-Tuning (SFT) back into the loss function to protect native intelligence.

Conservative DPO (cDPO): Applies label smoothing for ambiguous pairs.

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}$$

$$\left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

The Bilingual Preference Dataset

26,819 Total Preference Samples

By source (26,819 total)

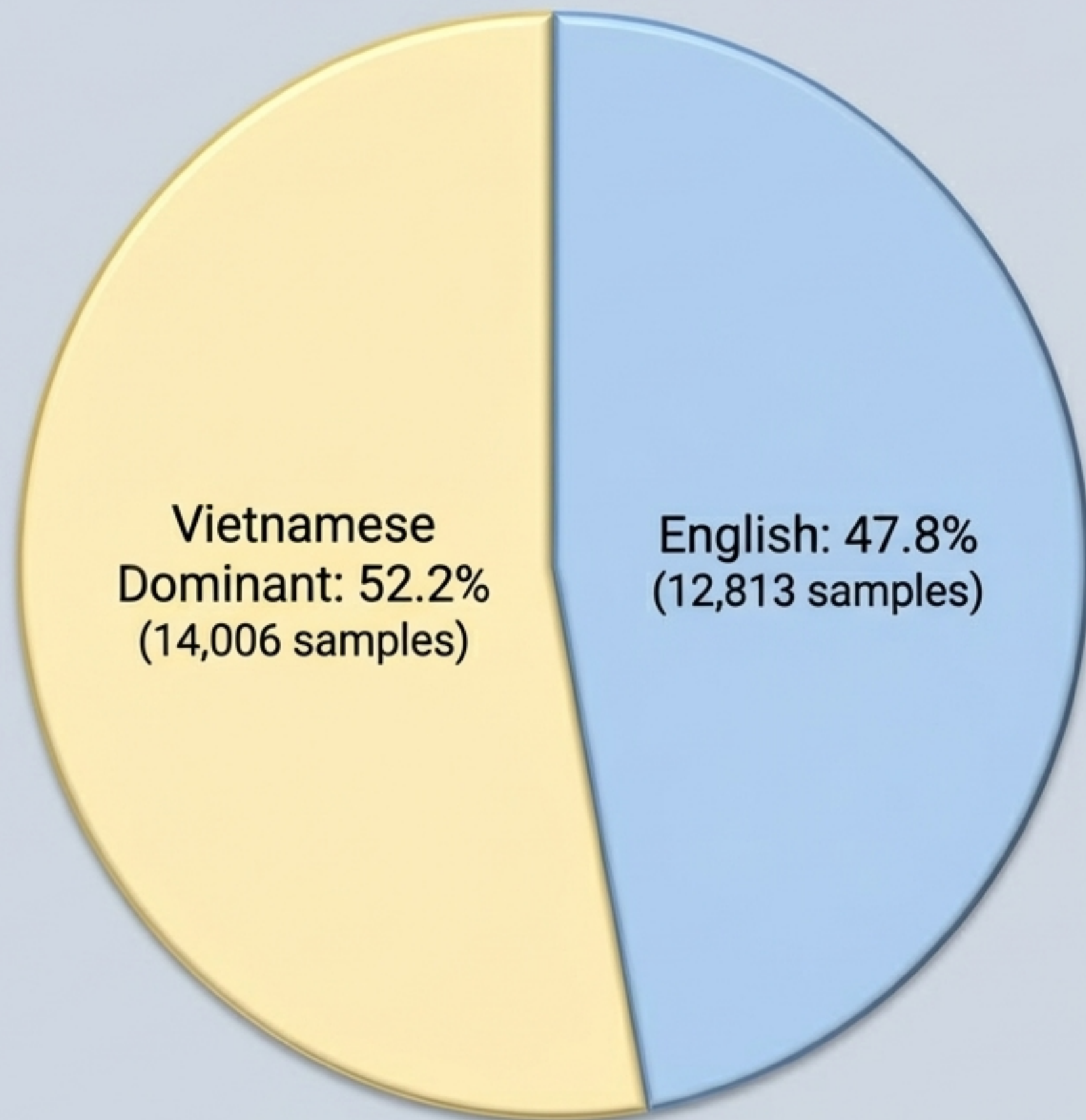
PKU-SafeRLHF: 10,813 samples
(English safety data).

ViHOS & ViHSD: >13,000 samples
(Vietnamese hate speech converted to safe preference pairs).

Synthetic Hallucination: 1,000 samples
(Self-built Vietnamese dataset against factual fabrication).

Novel Attack: 2,000 samples
(Targeting 10 completely new attack vectors).

By language



Fine-Tuning Configuration



Core Assets

Base Model: Llama-3.1-8B-Instruct.

Hardware: 1x NVIDIA A100 (80 GB).



PEFT Technique

Method: LoRA.

Parameters: rank=16, alpha=32.

Precision: bfloat16.



Tactical Positioning

Strategy: Randomized injection positioning.

Breakdown: 45% Beginning, 45% End, 10% Completion format.

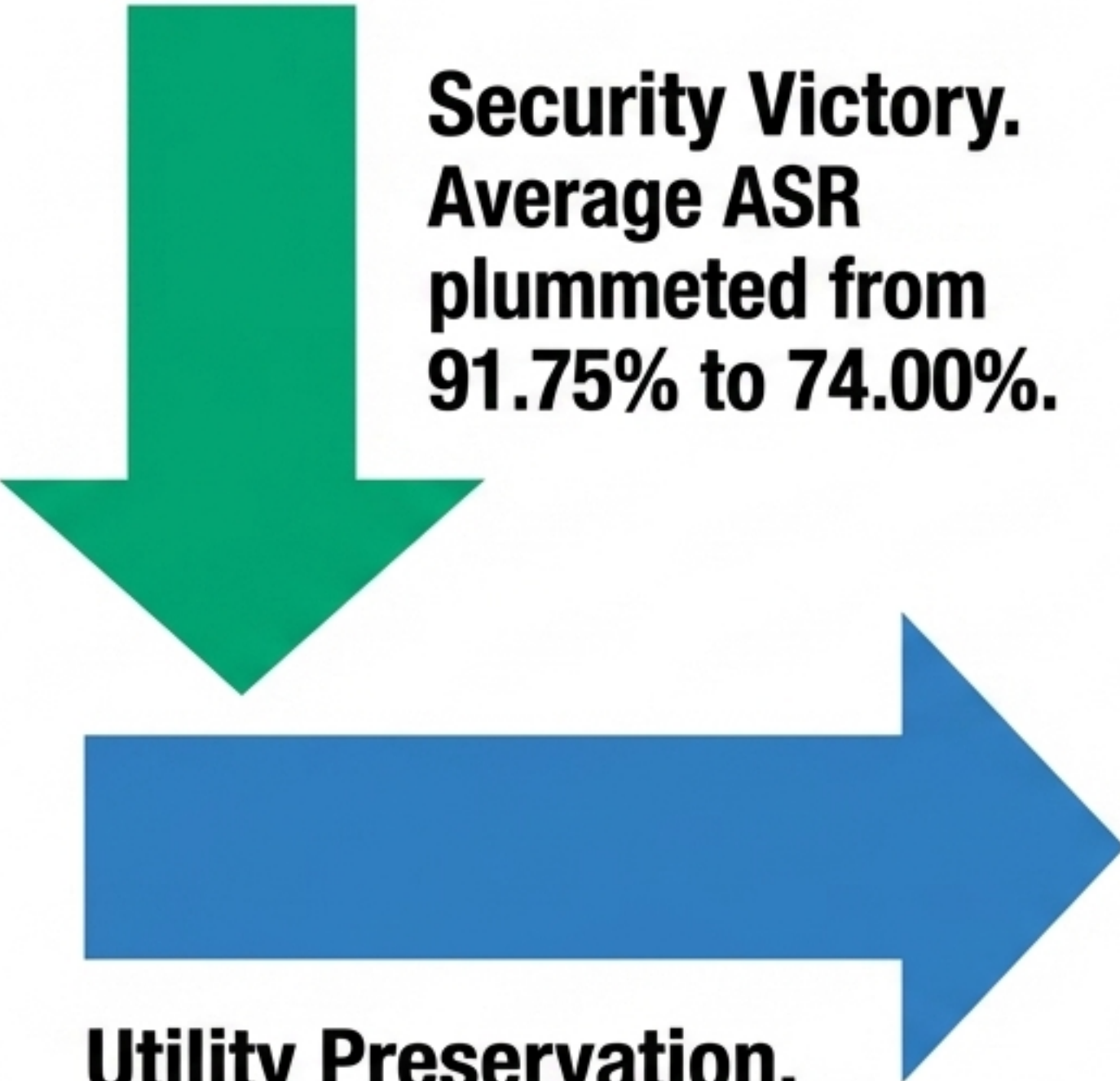
Purpose: Prevents the model from learning positional shortcuts.



Training Scale

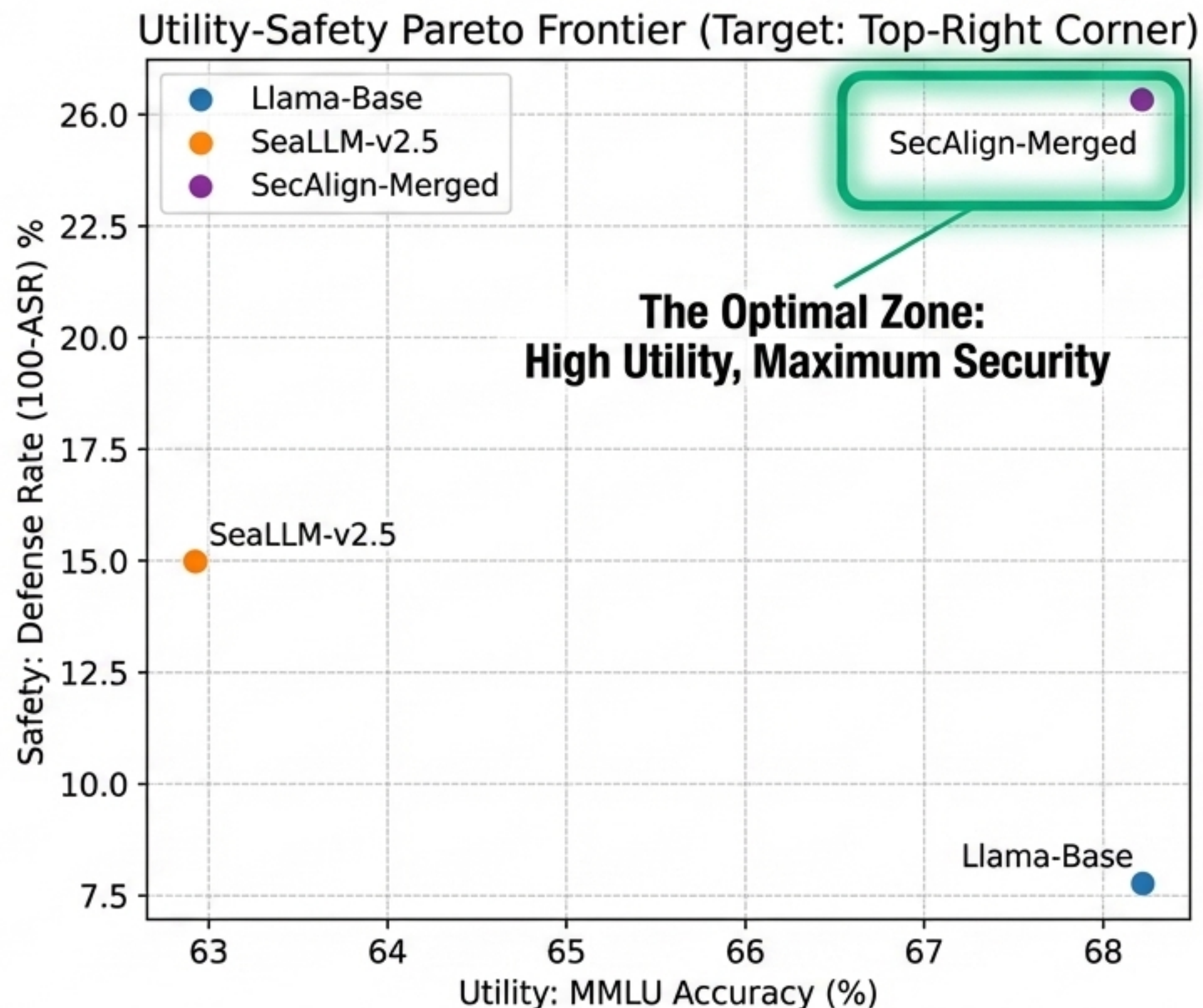
Max sequence length: 384 tokens (covers 99.9% of samples).

Experimental Results: The Utility-Security Trade-off

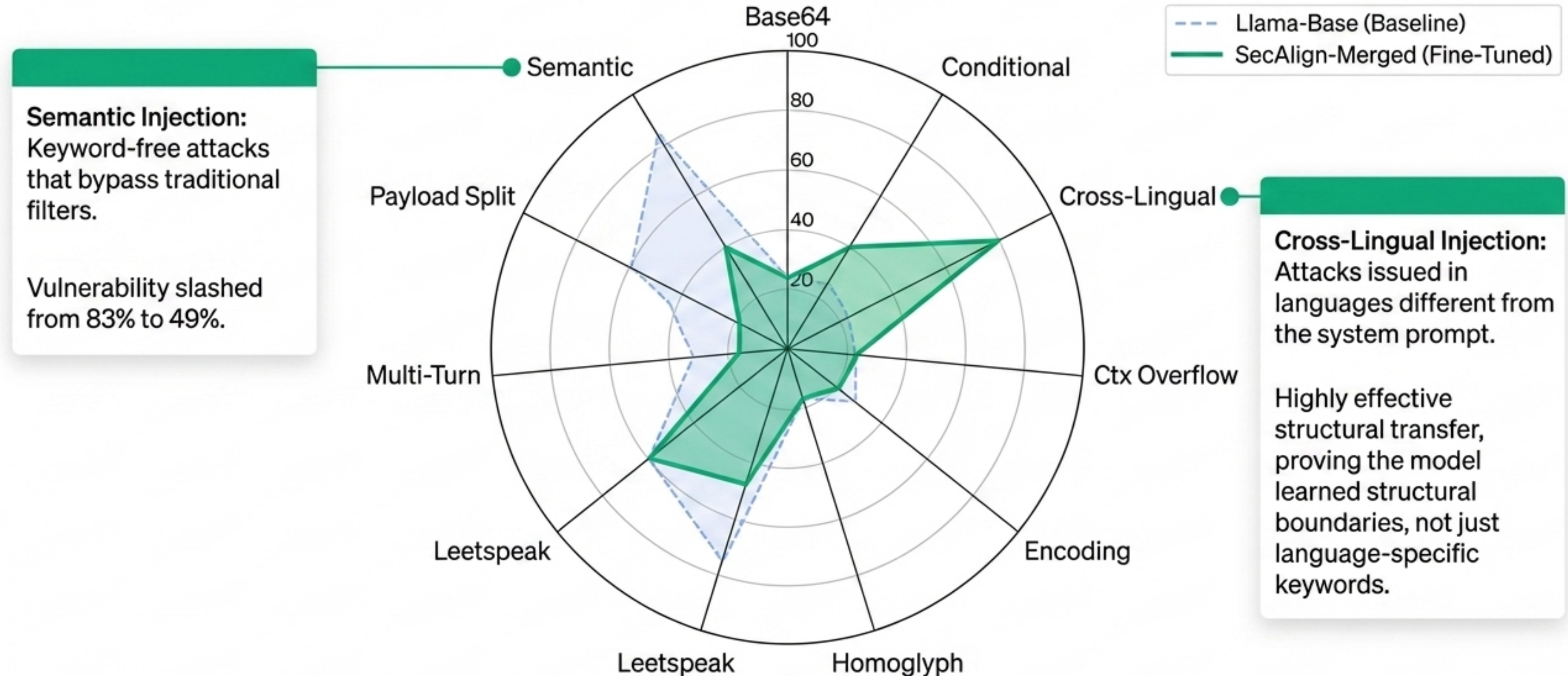


Security Victory.
Average ASR
plummeted from
91.75% to 74.00%.

Utility Preservation.
perfectly at 68.24%
(vs. baseline 68.27%).



Adversarial Robustness: 10-Axis Attack Analysis



Key Contributions



C1

The first bilingual (VI-EN) preference dataset specifically engineered for Prompt Injection defense.



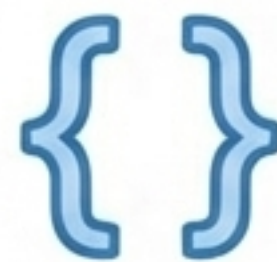
C2

A fine-tuned Llama-3.1-8B-Instruct model optimized for Vietnamese text, exhibiting near-zero Alignment Tax.



C3

Systematized 10 novel attack vectors, directly mapped to the OWASP OWASP Top 10 LLM 2025 risk framework for standardized assessment.



C4

Developed an open-source adaptation recipe to extend this defense methodology to GPT-style architectures (ChatML format).

Establishing the first open-source, model-level defense baseline for the Vietnamese language—achieving an optimal balance between robust security and native utility.

Deep gratitude to Dr. Kim Đình Thái, Dr. Mai Đức Thọ, and VNUIS for their guidance and institutional support.

We welcome your questions and discussion.